KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3113 | 3 Credits: 1.5 | Program Core | Source-grounded

Tool Engineering Drawing

 To provide a set of rules regarding the dimensions and tolerances of mating machined

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3113 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3113.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Tool & Die Engineering
Course code	3113
Course title	Tool Engineering Drawing
Semester	3 Credits: 1.5
Course category	Program Core
Periods per week	3 (L:0 T:0, P:3) Periods per semester: 45
Periods per semester	45
Credits	1.5
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

9 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide a set of rules regarding the dimensions and tolerances of mating machined
- To give the standard procedure of indicating surface texture and its application on
- To explain the forms and proportion of screw threads &fasteners
- To explain the importance of geometrical tolerance
- To prepare assembly views and to create the detail drawing of its elements

Course outcomes

CO	Official outcome	Study level
CO1	Demonstrate different type of Fasteners. 9 Applying Prepare dimensional drawings with tolerances and	See official syllabus
CO2	9 Applying standard procedure of surface texture. Apply the concept of sectional views to depict	See official syllabus
CO3	assembly views and detailed drawings of Machine 12 Applying elements. Sketch assembly views and detailed drawings of	See official syllabus
CO4	12 Applying Jig, Fixture & Dies.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3 3
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Demonstrate different type of Fasteners	3	As specified
Module 2	of surface texture	3	As specified
Module 3	detailed drawings of Machine elements.	2	As specified
Module 4	Sketch assembly views and detailed drawings of Jig, Fixture & Dies.	1	As specified

MODULE 1

Demonstrate different type of Fasteners

This module is presented from the official syllabus scope for Tool Engineering Drawing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate different type of Fasteners
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 3 Applying profiles.

3 Applying profiles. is an official topic in Module 1 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying profiles. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying profiles. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying profiles. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Applying profiles. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Sketch different types of bolts and nuts. 3 Applying Illustrate various types of rivet heads.

Sketch different types of bolts and nuts. 3 Applying Illustrate various types of rivet heads. is an official topic in Module 1 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Sketch different types of bolts and nuts. 3 Applying Illustrate various types of rivet heads. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sketch different types of bolts and nuts. 3 applying illustrate various types of rivet heads. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Sketch different types of bolts and nuts. 3 Applying Illustrate various types of rivet heads. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Sketch different types of bolts and nuts. 3 Applying Illustrate various types of rivet heads. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▼ 3 Applying Screw thread nomenclature. Major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Buttress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification &Conventional representation of threads. Bolt, nut, set screws, locking screws, locks nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints). Prepare dimensional drawings with tolerances and standard procedure

3 Applying Screw thread nomenclature. Major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Buttress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification &Conventional representation of threads. Bolt, nut, set screws, locking screws, locks nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints). Prepare dimensional drawings with tolerances and standard procedure is an official topic in Module 1 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Screw thread nomenclature. Major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Buttress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification &Conventional representation of threads. Bolt, nut, set screws, locking screws, locks nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints). Prepare dimensional drawings with tolerances and standard procedure using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying screw thread nomenclature. major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. forms of thread - metric thread, v thread, bsw thread, american national thread, buttress thread, square thread, acme thread, and worm thread. single and multi-start thread. thread specification &conventional representation of threads. bolt, nut, set screws, locking screws, locks nuts. socket head cap screw, allen screw, grub screw and csk screw and assembly, foundation bolts etc. draw riveted joints calculate the pitch and arrangements of rivet in rows (lap and butt joints). prepare dimensional drawings with tolerances and standard procedure should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Screw thread nomenclature. Major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Buttress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification &Conventional representation of threads. Bolt, nut, set screws, locking screws, locks nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints). Prepare dimensional drawings with tolerances and standard procedure means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Applying Screw thread nomenclature. Major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Buttress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification &Conventional representation of threads. Bolt, nut, set screws, locking screws, locks nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints). Prepare dimensional drawings with tolerances and standard procedure □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# of surface texture

This module is presented from the official syllabus scope for Tool Engineering Drawing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of surface texture
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Applying fits using standard tables 3 Interpret various Surface texture symbols

Applying fits using standard tables 3 Interpret various Surface texture symbols is an official topic in Module 2 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying fits using standard tables 3 Interpret various Surface texture symbols using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying fits using standard tables 3 interpret various surface texture symbols should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying fits using standard tables 3 Interpret various Surface texture symbols means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Applying fits using standard tables 3 Interpret various Surface texture symbols □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□

[illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 3 Applying from drawings

3 Applying from drawings is an official topic in Module 2 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying from drawings using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying from drawings should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying from drawings means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Applying from drawings □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ Apply Geometrical Tolerances in drawings 3 Applying Classifications of dimensions- functional, nonfunctional, auxiliary and features. IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances Basic size deviations and tolerance. Method of placing limit dimensions. – Bilateral, unilateral. Fits. – Clearance fit, transition and interference fit. General tolerance. – IS 2102{part1 and part2}. Compute fit and tolerance from tables. Introduction, - Types of surface texture, waviness nomenclature of surface texture. Indication of surface roughness. Symbols of specifying the direction of lay. Machining symbols. Introduction about geometrical tolerance. Tolerance frame and features Datum. – Primary, secondary and tertiary datum Apply the concept of sectional views to depict assembly views and

Apply Geometrical Tolerances in drawings 3 Applying Classifications of dimensions- functional, nonfunctional, auxiliary and features. IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances Basic size deviations and tolerance. Method of placing limit dimensions. – Bilateral, unilateral. Fits. – Clearance fit, transition and interference fit. General tolerance. – IS 2102{part1 and part2}. Compute fit and tolerance from tables. Introduction, - Types of surface texture, waviness nomenclature of surface texture. Indication of surface roughness. Symbols of specifying the direction of lay. Machining symbols. Introduction about geometrical tolerance. Tolerance frame and features Datum. – Primary, secondary and tertiary datum Apply the concept of sectional views to depict assembly views and is an official topic in Module 2 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply Geometrical Tolerances in drawings 3 Applying Classifications of dimensions- functional, nonfunctional, auxiliary and features. IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances Basic size deviations and tolerance. Method of placing limit dimensions. – Bilateral, unilateral. Fits. – Clearance fit, transition and interference fit. General tolerance. – IS 2102{part1 and part2}. Compute fit and tolerance from tables. Introduction, - Types of surface texture, waviness nomenclature of surface texture. Indication of surface roughness. Symbols of specifying the direction of lay. Machining symbols. Introduction about geometrical tolerance. Tolerance frame and features Datum. – Primary, secondary and tertiary datum Apply the concept of sectional views to depict assembly views and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply geometrical tolerances in drawings 3 applying classifications of dimensions- functional, nonfunctional, auxiliary and features. is specified tolerance. – indian standard/iso system of limits and fits (is919, iso 286) – allowances basic size deviations and tolerance. method of placing limit dimensions. – bilateral, unilateral. fits. – clearance fit, transition and interference fit. general tolerance. – is 2102{part1 and part2}. compute fit and tolerance from tables. introduction, - types of surface texture, waviness nomenclature of surface texture. indication of surface roughness. symbols of specifying the direction of lay. machining symbols. introduction about geometrical tolerance. tolerance frame and features datum. – primary, secondary and tertiary datum apply the concept of sectional views to depict assembly views and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply Geometrical Tolerances in drawings 3 Applying Classifications of dimensions- functional, nonfunctional, auxiliary and features. IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances Basic size deviations and tolerance. Method of placing limit dimensions. – Bilateral, unilateral. Fits. – Clearance fit, transition and interference fit. General tolerance. – IS 2102{part1 and part2}. Compute fit and tolerance from tables. Introduction, - Types of surface texture, waviness nomenclature of surface texture. Indication of surface roughness. Symbols of specifying the direction of lay. Machining symbols. Introduction about geometrical tolerance. Tolerance frame and features Datum. – Primary, secondary and tertiary datum Apply the concept of sectional views to depict assembly views and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-▯▯ Apply Geometrical Tolerances in drawings 3 Applying Classifications of dimensions- functional, nonfunctional, auxiliary and features. IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances Basic size deviations and tolerance. Method of placing limit dimensions. – Bilateral, unilateral. Fits. – Clearance fit, transition and interference fit. General tolerance. – IS 2102{part1 and part2}. Compute fit and tolerance from tables. Introduction, - Types of surface texture, waviness nomenclature of surface texture. Indication of surface roughness. Symbols of specifying the direction of lay. Machining symbols. Introduction about geometrical tolerance. Tolerance frame and features Datum. – Primary, secondary and tertiary datum Apply the concept of sectional views to depict assembly views and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

detailed drawings of Machine elements.

This module is presented from the official syllabus scope for Tool Engineering Drawing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: detailed drawings of Machine elements.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

▼ 3 Applying plan of various machine parts.

3 Applying plan of various machine parts. is an official topic in Module 3 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying plan of various machine parts. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying plan of various machine parts. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying plan of various machine parts. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Applying plan of various machine parts. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Introduce assembly and part drawing 9 Applying Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. Screw Jack 2. C Clamp 3. Tool Makers clamp 4. Lathe Tailstock 5. Lathe spindle 6. Bench Vice etc.

Introduce assembly and part drawing 9 Applying Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. Screw Jack 2. C Clamp 3. Tool Makers clamp 4. Lathe Tailstock 5. Lathe spindle 6. Bench Vice etc. is an official topic in Module 3 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduce assembly and part drawing 9 Applying Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. Screw Jack 2. C Clamp 3. Tool Makers clamp 4. Lathe Tailstock 5. Lathe spindle 6. Bench Vice etc. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduce assembly and part drawing 9 applying review of sectioning – conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining general rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. screw jack 2. c clamp 3. tool makers clamp 4. lathe tailstock 5. lathe spindle 6. bench vice etc. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduce assembly and part drawing 9 Applying Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. Screw Jack 2. C Clamp 3. Tool Makers clamp 4. Lathe Tailstock 5. Lathe spindle 6. Bench Vice etc. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Introduce assembly and part drawing 9 Applying Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. Screw Jack 2. C Clamp 3. Tool Makers clamp 4. Lathe Tailstock 5. Lathe spindle 6. Bench Vice etc. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▸ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Sketch assembly views and detailed drawings of Jig, Fixture & Dies.

This module is presented from the official syllabus scope for Tool Engineering Drawing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Sketch assembly views and detailed drawings of Jig, Fixture & Dies.
- Official duration: As specified
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

▼ 12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs & Fixtures - P H Joshi Online Resources: Sl.No Website Link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf

12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs & Fixtures - P H Joshi Online Resources: Sl.No Website Link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf is an official topic in Module 4 of Tool Engineering Drawing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs & Fixtures - P H Joshi Online Resources: Sl.No Website Link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 12 applying moulds with gd & t, surface roughness values and bill of materials. following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. drill jig – post & latch type welding fixture, milling fixture, indexing fixture press tool assembly – blanking tool plastic moulding tool assembly – injection moulding. text/ reference: t/r book title/author t1 machine drawing - p i varghese and k.c john r1 engineering drawing - n.d. bhatt r2 jigs and fixtures – hiram.e.grant r3 jigs and fixtures design manual –p h joshi r4 jigs & fixtures - p h joshi online resources: sl.no website link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs & Fixtures - P H Joshi Online Resources: Sl.No Website Link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs &

Fixtures - P H Joshi Online Resources: Sl.No Website Link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps → Application →
Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure →
Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 3 Applying profiles.. (3 marks)**

▶ **Define or explain Sketch different types of bolts and nuts. 3 Applying Illustrate various types of rivet heads.. (3 marks)**

▶ **Define or explain 3 Applying Screw thread nomenclature. Major die, minor dia, pitch, lead, flank, crest, root, thread angle etc. Forms of thread - metric thread, V thread, BSW thread, American national thread, Buttress thread, Square thread, ACME thread, and worm thread. Single and multi-start thread. Thread specification &Conventional representation of threads. Bolt, nut, set screws, locking screws, locks nuts. Socket head cap screw, Allen screw, grub screw and CSK screw and assembly, foundation bolts etc. Draw riveted joints Calculate the pitch and arrangements of rivet in rows (Lap and Butt joints). Prepare dimensional drawings with tolerances and standard procedure. (3 marks)**

Module 2

▶ **Define or explain Applying fits using standard tables 3 Interpret various Surface texture symbols. (3 marks)**

▶ **Define or explain 3 Applying from drawings. (3 marks)**

▶ **Define or explain Apply Geometrical Tolerances in drawings 3 Applying Classifications of dimensions- functional, nonfunctional, auxiliary and features. IS specified tolerance. – Indian standard/ISO system of limits and fits (IS919, ISO 286) – Allowances Basic size deviations and tolerance. Method of placing limit dimensions. – Bilateral, unilateral. Fits. – Clearance fit, transition and interference fit. General tolerance. – IS 2102{part1 and part2}. Compute fit and tolerance from tables. Introduction, - Types of surface texture, waviness nomenclature of surface texture. Indication of surface roughness. Symbols of specifying the direction of lay. Machining symbols. Introduction about geometrical tolerance. Tolerance frame and features Datum. – Primary, secondary and tertiary datum Apply the concept of sectional views to depict assembly views and. (3 marks)**

Module 3

▶ **Define or explain 3 Applying plan of various machine parts.. (3 marks)**

▶ **Define or explain Introduce assembly and part drawing 9 Applying Review of sectioning – Conventions showing the section – symbolic representation of cutting plane- types of section – full section, half section, offset section, revolved section, broken section, removed section – section lining General rules of sectioning. - hatching of single object, adjacent object, thin material, large area, rib & web section. Detailed drawings of following machine parts are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. 1. Screw Jack 2. C Clamp 3. Tool Makers clamp 4. Lathe Tailstock 5. Lathe spindle 6. Bench Vice etc.. (3 marks)**

Module 4

▶ **Define or explain 12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs & Fixtures - P H Joshi Online Resources: Sl.No Website Link 1 http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Applying profiles.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Applying fits using standard tables 3 Interpret various Surface texture symbols**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Applying plan of various machine parts.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **12 Applying Moulds with GD & T, surface roughness values and bill of materials. Following detailed drawings are given to students to assemble and draw the sectional or plain elevations / plans / and side views with dimensioning and bill of materials. Drill Jig – post & latch type Welding fixture, milling fixture, indexing fixture Press tool assembly – Blanking tool Plastic Moulding tool assembly – Injection moulding. Text/ Reference: T/R Book Title/Author T1 Machine drawing - P I Varghese and K.C John R1 Engineering Drawing - N.D. Bhatt R2 Jigs and Fixtures – Hiram.E.Grant R3 Jigs and Fixtures Design Manual –P H Joshi R4 Jigs & Fixtures - P H Joshi Online Resources: Sl.No Website Link 1**

http://www.scce.ac.in/noticeboard/10367_23062 complete-manual.pdf 2 http://www.ignou.ac.in/upload/jig.pdf.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- http://www.scce.ac.in/noticeboard/10367_23062016pdp-complete-manual.pdf http://www.ignou.ac.in/upload/jig.pdf

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3113. Verify institutional updates against the current official curriculum.